

Production Target Optimization

Pareto BO over the Stickman PS target — methodology

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Mu2e — autoresearch / closed-loop BO

Status (2026-06-11): 39 evals on disk (pt001, pt002, ptX01, ptX02, ptX05). **5 Stickman-dominators** so far; best simultaneous beat is `ptX05R02_07` at $\mu=2.313\times 10^{-3}$ (+6.8%), dose= 1.387×10^{-9} Gy/POT (−8.7%). Best μ -only `ptX05R02_04` 2.373×10^{-3} (+9.6%); best dose-only `ptX01R00_00` 8.03×10^{-10} Gy/POT (−47%). See `prodtarget6d_talk` for the parallel 6D variant.

What we optimize

Baseline: **MDC2025aq Stickman v1.0** — 35 Inconel-718 plates,
`r0ut=3.15 mm` , `t=5 mm` , rotated 14° about Y, in the PS vacuum.

10-D + 1 integer search space — three knots of a **quadratic profile**
along the stack for `r0ut` , `plateThickness` , `plateLugThickness` , plus
plate count:

knob (3 each: upstream / mid / downstream)	range	units
<code>r0</code> , <code>r1</code> , <code>r2</code> — <code>r0ut</code> profile	2.0 – 4.5	mm
<code>t0</code> , <code>t1</code> , <code>t2</code> — plate thickness	3.0 – 8.0	mm
<code>l0</code> , <code>l1</code> , <code>l2</code> — plate lug thickness	4.0 – 12.0	mm
<code>N</code> — number of plates (Integer)	25 – 45	—

Smooth, buildable, spans the full envelope around Stickman (`r=3.15` , `t=5` , `l=8`).

Two competing objectives

- **Maximize** `mu_per_POT` — stopped muons per proton on target.
- **Minimize** `peak_dose_Gy_per_POT` = `max_i (Edep_i / mass_i)` over the 35 plates, with

$$\text{mass}_i = \pi \cdot r_{\text{Out}_i}^2 \cdot t_i \cdot \rho, \quad \rho_{\{\text{Inconel718}\}} = 8.19 \text{ g/cm}^3.$$

Per-plate Edep via a custom `StepPointMC` collection per `ProductionTargetPlate%02d` volume.

Sanity: $1.5\text{e-}9 \text{ Gy/POT} \times 6 \times 10^{15} \text{ POT/yr} \approx \mathbf{9 \text{ MGy/yr}}$ — same order as Stickman published design point.

Noise floor: `pt001` vs `pt002` (replicas) agree to **<0.5%** on peak dose.

Why peak specific dose, not stack-total Edep

Stack-total Edep mis-ranks the thermally stressed designs three ways:

- 1. **1/rOut² coupling is invisible** — shrinking rOut keeps Edep roughly constant but divides by a smaller mass, so peak dose explodes.
- 2. **Hot-plate masking** — one plate at melt + 34 cool ≈ uniform stack on the total. Operationally fatal, statistically silent.
- 3. **N-scaling artifact** — more plates = more total, even when each plate sees less dose.

Empirical confirmation (from our 4 evals):

config	stack Edep [MeV/POT]	peak [Gy/POT]	rank by Edep	rank by peak
pt001 (baseline)	422	1.53e-9	1	2
ptX01R00_00	412	8.0e-10 (best)	2	1
ptX01R00_01	179 (lowest!)	4.0e-9 (worst, 2.6× base)	4 (looks safest)	4

Lowest-Edep row = **highest-dose** row. Pareto BO on stack-total chases the corner; peak dose corrects it.

The optimizer: qLogNEHVI

Same picker stack as the foils campaign — proven on `(S/√B, -log calo)`,
transplanted onto `(mu_per_POT, -log10 peak_dose)`.

- **Multi-objective** — maximizes Pareto hypervolume; delivers the *whole* dose-vs-yield front in one campaign.
- **Noisy** — marginalizes G4 per-POT Poisson noise (~0.5% on peak dose).
- **Log-stabilized** — fixes vanishing-gradient failure of plain qNEHVI near saturation.

`q=4` per round, sequential-greedy via batched fantasies (joint mode blew past 10-min wall at `q=10` on foils).

Closed-loop architecture is shared — `prodtarget` is one entry in `MODE_SPECS` + one Y-column branch in `botorch_predict._load_history_tensor`; no per-mode picker code.

GP cloud — current 10-eval landscape

65 536 Sobol-sampled posterior means; GP trained on **n=10**.

- **Champion** **ptX02R00_00** sits up-and-left of Stickman: +3% μ , -18% dose.
- Predicted μ ceiling now reaches 2.14×10^{-3} (up from 1.99 at $n=4$); GP has learned the upper envelope.
- Picker recently explored a high-dose ridge (**R01_02** at 3.3×10^{-9}) — qLogNEHVI buying hypervolume, not just chasing the knee.

Status & next steps

Now (39 evals): 4 baseline-era rows + ptX01 (2) + ptX02 (7) + ptX05 (28, $q=10 \times 3$ rounds, qNEHVI). 5 Stickman-dominators identified, biggest simultaneous beat `ptX05R02_07` at $\mu=2.313e-3$ / dose= $1.387e-9$.

Pareto leaders (peak dose $\uparrow$): `ptX01R00_00` ($8.0e-10$ / $1.98e-3$) $\rightarrow$ `ptX05R00_04` ($8.7e-10$ / 2.11) $\rightarrow$ `ptX02R00_03` ($1.16e-9$ / 1.82) $\rightarrow$ `ptX05R00_05` ($1.28e-9$ / 2.19) $\rightarrow$ **`ptX05R02_07` ($1.39e-9$ / 2.31)** $\rightarrow$ `ptX05R02_04` ($1.84e-9$ / 2.37 — best μ).

Next: parallel 6D variant under separate deck (`prodtarget6d_talk`); 10D campaign next move TBD (likely ptX06 to push further into the dose-favorable corner around `ptX05R00_04` at 8.7×10^{-10} / 2.11×10^{-3}).

Deferred:

- **Path B NIEL** as 3rd objective — custom SD still missing neutron channel; ICRU49-only underestimates DPA $\sim 10\times$ vs published Stickman 10 DPA/yr.
- `plateMaterial` categorical (Inconel/W/Ta) — needs mixed-model (qParEGO / one-hot) rather than continuous qNEHVI.
- Closed-loop final-round orphan-on-exit bug — `ptX02R01_03` lost when parent exited clean at `max_rounds`; manual harvest TBD.